

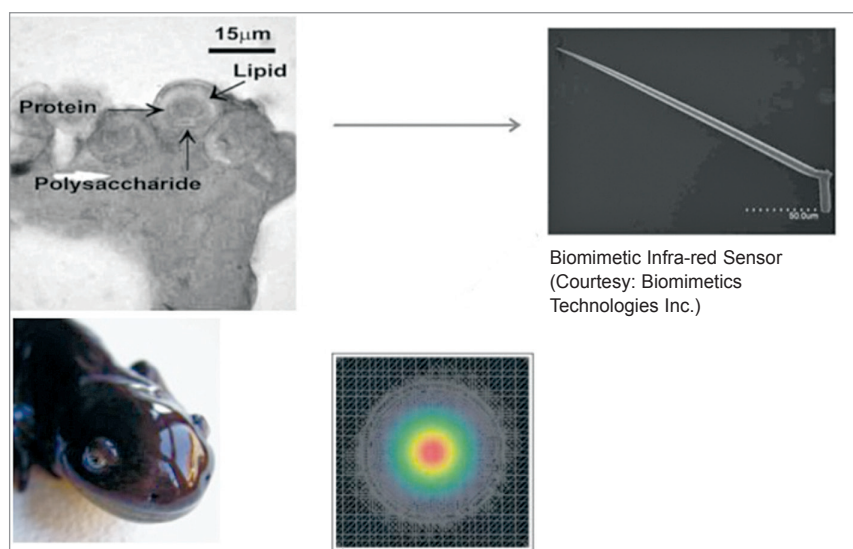


Fig. 11: Biomimetic-inspired infra-red sensor



Fig. 12: Biomimetic membrane (Courtesy: Applied Biomimetic)

aspects of practical life. MiRoCLOUD, created by Consequential Robotics, a new browser based platform, provides the opportunity for individuals of all ages and aptitudes to understand and develop the coding skills. MiRo keeps students from primary to higher education curiously engaged as it provides scope for creativity with fifty different variables making limitless opportunities for coding. MiRo develops competency, critical thinking, and free-ranging discussion in STEM that integrates the four areas of science, technology, engineering, and mathematics to provide a multi-level learning experience.

MiRo works in two ways—the cloud based and the installation based. MiRoCODE starts coding with Blockly that segues to Python. Coding is made more attractive and amusing by MiRo-E (Fig. 9), a biomimetic robot, developed by MiRoCLOUD in association with universities and educationalists. Sebastian Conran Associates studio has changed the perspective of viewing a robot merely

as a gadget to a lovable friend full of potential. Animals have different forms of physical, emotional, and social intelligence, which inspired the MiRo team to design MiRo-E robot as an animal-like structure than human-like structure.

MiRo-E robot comes with a wide-ranging suite of biomimetic sensors including stereo vision, location hearing, ultrasonic ranging, light level sensors, infrared cliff sensors, tactile sensors on the body and head, and interceptive sensors such as twin accelerometers, joint position sensors (proprioception), temperature, and battery level sensors. There are eleven degrees of freedom—tail wagging and drooping, blinking, ear-rotation, head movements, coloured lights—to make the communication more expressive so that children ranging from normal to special needs can communicate their wants, thoughts, and opinions.

Some leading biomimetic companies and their products

Biomimetics finds applications in robotics, nanotechnology, medical industry, artificial intelligence (AI), military, and much more.

Avinent manufactures digital dental implant systems (Fig. 10) that provide new and advanced products

for the medical sector. Their products are inspired by the biochemical procedures in nature resulting in biomimetic advanced surface (BAS). The CAD CAM division in Avinent provides personalised prosthesis solutions.

BioHorizons is developing biomimetic-inspired infra-red sensors (Fig. 11), biomimetic magnetic chip, etc. The optics in their devices utilises salamander's eye models. Their products find application in breast cancer detection, endoscopy, image reconstruction, magnetic sensor, and surgical instrumentation.

Applied Biomimetic is developing high-performance permeable membrane (Fig. 12) with narrow molecular size exclusion cut-off. The concept that inspired them is the nature's ability to separate chemical elements at the molecular level using highly specialised proteins as tools. This biomimetic membrane finds applications in water, life science, enzyme, food, and dairy separation.

Conclusion

A product designed using biomimetic technology is not only innovative but also potentially sustainable. Nature works as a system that gives effective and sustainable outcomes. Researchers from various fields are learning this principle and creating potential, productive, and sustainable products. The main characteristic of the industrial revolution is 'dominating and exploiting the nature.' In contrast, biomimetics innovations are characterised by 'learning from nature and mimicking the actual biological structures and functions.' The rigid environmental policies in protecting the environment have powered the development of biomimetics in many countries.

The biomimetic market is endorsing excellent growth. So, let us revolutionise our inventions by drawing encouragement from nature and hypothesize designs that create effective sustainable materials. **EFY**



Dr. Krishna Naga is a physics teacher with over twenty years of experience in this field. Her hobbies include reading and writing